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Internal loads in the human tibia during gait

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ABSTRACT

Background: Internal loads in long bones are of special interest, for testing and optimization of current implant designs for fracture treatment. Inverse dynamic musculoskeletal models were successfully used to determine the muscle forces in the lower extremities, however the internal loads expressed as forces and moments along the human tibia during gait have not been reported.

Methods: A musculoskeletal model of the lower extremities, developed and provided under public domain from the AnyBody research group (www.anybody.aau.dk), was modified to determine the three-dimensional internal loads along the tibial axis during gait. To corroborate this numerical model, the calculated resultant hip contact force as well as the axial force on the tibial plateau were compared to in vivo data from the literature.

Findings: The highest internal loads were the axial force with up to 4.7 bodyweight and the bending moment in the sagittal plane with up to 71.6 bodyweight times millimetre in the late stance phase. The extreme values of the internal loads along the tibial axis varied up to 1.5 bodyweight and 85.7 bodyweight times millimetre. In the distal part of the tibia, the axial force dominated the three-dimensional internal load case, whereas the internal moments became more significant with increasing distance from the ankle joint.

Interpretation: This study provides for the first time the three-dimensional internal loads, expressed as forces and moments along the human tibia during normal gait. The results of this study could be used to improve the mechanical behaviour of current implant designs for the treatment of tibial fractures to avoid implant failures under in vivo loading conditions.

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1. Introduction

Internal loads acting in vivo in long bones under daily activities are of special interest in fracture healing research. In vitro tests of fixation devices mounted on bone fragments are often used to analyze the primary as well as the long term stability of the bone–implant complex (BIC) (Ahmad et al., 2007; Stoffel et al., 2007). To decide whether the in vitro determined load to failure of the tested BIC is critical, the internal loads have to be known.

Telemetrizised implants were successfully used to determine the internal loads acting inside the implant during fracture healing based on the deformations of the fixation device (Schneider et al., 2001; Seide et al., 2004). Often, the fracture site could carry some of the load, which leads to smaller deformations in this device and therefore to an underestimation of the internal loads. The effect of this load sharing mechanism is amplified in the late healing stage due to the newly formed bone in the healing region. Due to this underestimation, which is difficult to quantify, measured loads are always patient specific and therefore not suitable for describing the internal loads in general.

Beside these in vivo measurements, inverse dynamic models were used to calculate the musculoskeletal loads during daily activities (Brand et al., 1994; Glitsch and Baumann, 1997; Heller et al., 2001; Stansfield et al., 2003; Taylor et al., 2004). Often, the musculoskeletal loads are expressed by the calculated muscle forces and joint reaction loads, which were used to apply physiological-like loading conditions on Finite-Element (FE) models (Duda et al., 1998; Duda et al., 2001; Duda et al., 2002; Gomez-Benito et al., 2007). For performing in vitro tests and discussing the results thereof, there is a need to know the internal loads, expressed as internal forces and moments. However, there is no information about these forces and moments along the human tibia during gait in the literature.

The development of a numerical musculoskeletal model is very complex and time consuming. Therefore we modified a muskuloskeletal model of the human lower extremities, which was developed and provided under public domain by the AnyBody Research Group (repository 6, www.anybody.aau.dk). The aim of our study was to determine the internal forces and moments along the longitudinal axis of the human tibia during gait.

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2. Methods

The musculoskeletal model used consists of seven rigid bodies representing the bones as well as the surrounding tissues of the lower human extremities, i.e. pelvis, thigh, shank and foot of both legs (Fig. 1). The hip joints were modeled as spherical joints with three rotational degrees of freedom (DoF). Both knee joints were simulated as hinge joints with one rotational DoF whereas the ankle joints had two rotational DoF according to a cardan joint. There were 35 Hill type muscles (Winters, 1990) for each leg. The parameters to describe the muscles behaviour were based on the data of S.L. Delp, provided by the International Society of Biomechanics (http://isbweb.org/data).

The model is driven by the gait analysis dataset "man" of Vaughan et al. (1992), based on the movement of 15 optical markers and the ground reaction force (GRF) measured with two Kistler plates. To estimate the internal loads, the rigid body of the shank was divided into two new rigid bodies which were combined with a rigid joint in-between. The cutting plane was always set perpendicular to the longitudinal axis with the distance L_d to the ankle joint (Fig. 1). Two new local, body fixed coordinate systems were defined parallel to the local coordinate system of the old rigid body. The origins of the new systems were placed in the geometrical middle of each segment on the longitudinal axis of the shank.

The new centres of mass (COM) were defined at the origin of the new local coordinate systems. The proximal mass (m_p) and the distal mass (m_d) were calculated based on the length of the position vectors ${}^p\underline{r}$ and ${}^d\underline{r}$ in such a way that the resulting COM of both bodies is located at the position of the original COM of the old rigid body

$$m_p = m\frac{|{}^d\underline{r}|}{|{}^p\underline{r}| + |{}^d\underline{r}|} \tag{1}$$

$$m_d = m\frac{|{}^p\underline{r}|}{|{}^p\underline{r}| + |{}^d\underline{r}|} \tag{2}$$

The inertia tensors ($\underline{\underline{J}}_p$ and $\underline{\underline{J}}_d$) of the new bodies were calculated in relation to the corresponding new local coordinate system based on the parallel-axes theorem (Young and Budynas, 2002). $\underline{\underline{J}}$ and L were the inertia tensor and the length of the old shank, respectively, and ${}^p r_y$ as well as ${}^d r_y$ the y-components of the position vectors

$$\underline{\underline{J}}_p = \begin{pmatrix} \frac{m_p(L-L_d)^2}{m_d L_d^2 + m_p(L-L_d)^2} & 0 & 0 \\ 0 & \frac{m_p^2 L_d}{m_d^2(L-L_d)+m_p^2 L_d} & 0 \\ 0 & 0 & \frac{m_p(L-L_d)^2}{m_d L_d^2 + m_p(L-L_d)^2} \end{pmatrix} \underline{\underline{J}}_{red} \tag{3}$$

$$\underline{\underline{J}}_d = \begin{pmatrix} \frac{m_d L_d^2}{m_d L_d^2 + m_p(L-L_d)^2} & 0 & 0 \\ 0 & \frac{m_d^2(L-L_d)}{m_d^2(L-L_d)+m_p^2 L_d} & 0 \\ 0 & 0 & \frac{m_d L_d^2}{m_d L_d^2 + m_p(L-L_d)^2} \end{pmatrix} \underline{\underline{J}}_{red}$$

$$\underline{\underline{J}}_{red} = \underline{\underline{J}} - \begin{pmatrix} m_p {}^p r_y^2 + m_d {}^d r_y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & m_p {}^p r_y^2 + m_d {}^d r_y^2 \end{pmatrix} \tag{4}$$

The muscle forces were calculated to fulfil the equlibrium conditions of the rigid body system in each simulated time step during gait by the AnyBody software (www.anybodytech.com). In total, 100 time steps were simulated. The redundancy problem of the muscle forces was solved through an optimization algorithm, with the maximal muscle activity as the objective function which had to be minimized. To corroborate the musculoskeletal model, the resultant hip contact force and the axial force on the tibial plateau were compared to in vivo measured data from the literature. The internal loads were determined through the reaction forces and moments of the rigid joint during one gait cycle. To obtain the distribution of the internal forces and moments along the tibial axis, the cutting plane (location of the rigid joint) was defined at nine different locations. The masses, inertia tensors and muscle attachments were defined according to the specific length of the rigid shank bodies.

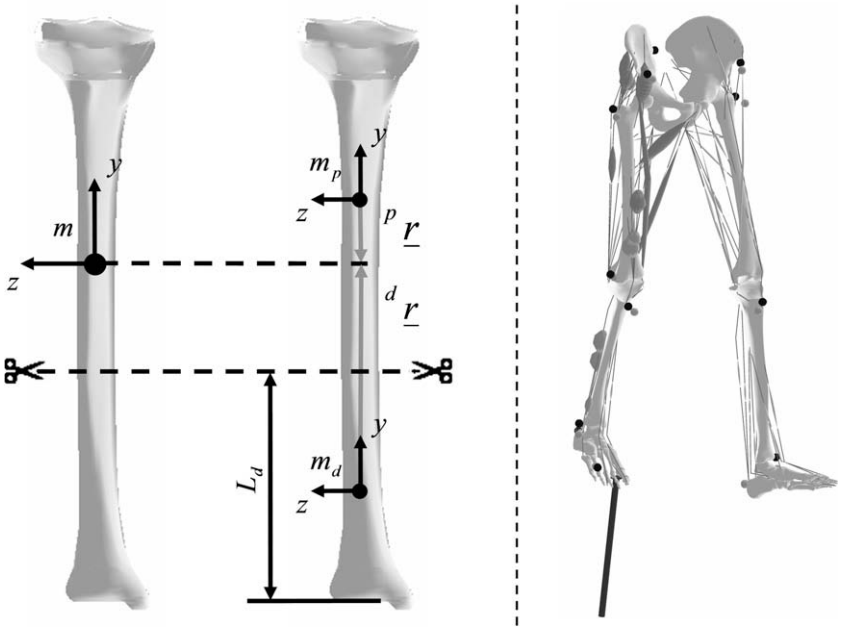


Fig. 1. Division of the right shank body with respect to the length of the distal part L_d (left). The position vectors in the new local coordinate systems ${}^p\underline{r}$ and ${}^d\underline{r}$ points to the centre of mass of the old shank body. Picture of the musculoskeletal model at $t_{norm} = 0.5$ (right).

3. Results

The calculated resultant hip contact force as well as the axial force on the tibial plateau showed loading peaks at the beginning ($t_{norm} \approx 0 \ldots 0.21$) and the end ($t_{norm} \approx 0.43 \ldots 0.64$) of the stance phase (Fig. 2). The peak values were 4.4 times bodyweight (BW) for the hip contact force and 3.3 BW for the knee contact force. The highest internal force in the tibia always acted in the longitudinal direction (F_y) during gait with the extreme value at the end of the stance phase (Fig. 3 and Supplementary data). The bending moments M_x and M_z in the tibial midshaft ($L_d = 0.5$ L) were of the same magnitude at the beginning of the stance phase but the moment in the sagittal plane displayed much higher values at the end of the stance phase. The extreme values of the internal loads along the tibial axis varied up to 1.5 BW and 85.7 BWmm (Fig. 3 and Supplementary data). In the distal region of the tibia, the axial force dominated the three-dimensional internal load case. However, with increasing distance from the ankle joint, the moments increased up to 71.6 BWmm in the sagittal plane.

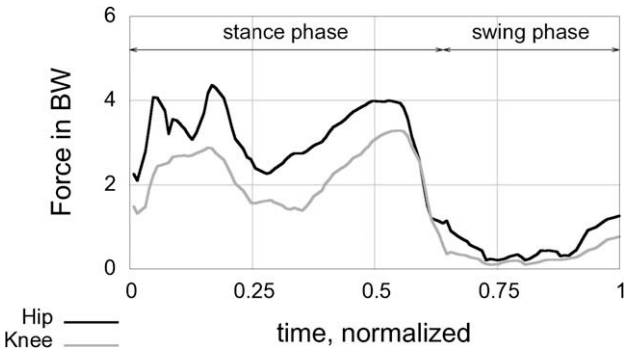


Fig. 2. Calculated resultant hip contact force and knee contact force (axial force on the tibial plateau) in bodyweight (BW) over the gait cycle.

4. Discussion

In this study we successfully modified an existing musculoskeletal model to determine the internal forces and moments in the human tibia during gait. These internal loads were strongly dependent on the time point of the gait cycle as well as the location along the tibial axis. The highest internal loads occurred in the late stance phase. At this time point of the gait cycle, the centre of pressure was located in the medio-ventral position of the foot and the GRF was in the medial, dorsal and proximal directions. Therefore, the GRF induced moments in the sagittal and frontal plane due to the lever arm between the action line of the GRF and the ankle joint. Both moments were captured through the shank muscles, especially by the Musculus Soleus ($F_{max} = 2.0$ BW) and Musculus Gastrocnemius ($F_{max} = 1.1$ BW), which led to the high internal loads in the late stance phase.

These muscle contractions combined with the GRF lead to a very high internal force in the axial direction (up to 4.7 BW), which might be critical for fixation devices, especially in the distal part of the tibia. Whereas, in the proximal part of the tibia the internal moments were much higher, which might lead to a failure of the bone–implant complex due to combined loading. In general, the patient tries to prevent the overloading of the treated leg and therefore to reduce the internal loads in the tibia during the early healing phase. However, the patient can not suppress all muscle contractions or prevent the overloading of the treated leg when walking with crutches without the correct expertise.

The limitations of our study were the basic limitations associated with numerical inverse dynamic musculoskeletal models, i.e. the calculated musculoskeletal loads were strongly dependent on factors such as the kinematics and external loads, the action lines of the muscles, the description of the muscle behaviour and the objective function of the optimization procedure (Brand et al., 1986; Duda et al., 1996; Hatze, 2002; Raikova and Prilutsky, 2001; Rasmussen et al., 2001). Most of these uncertainties are difficult to quantifiy, therefore the resultant hip contact force as well as the axial force on the tibial plateau were compared to in vivo

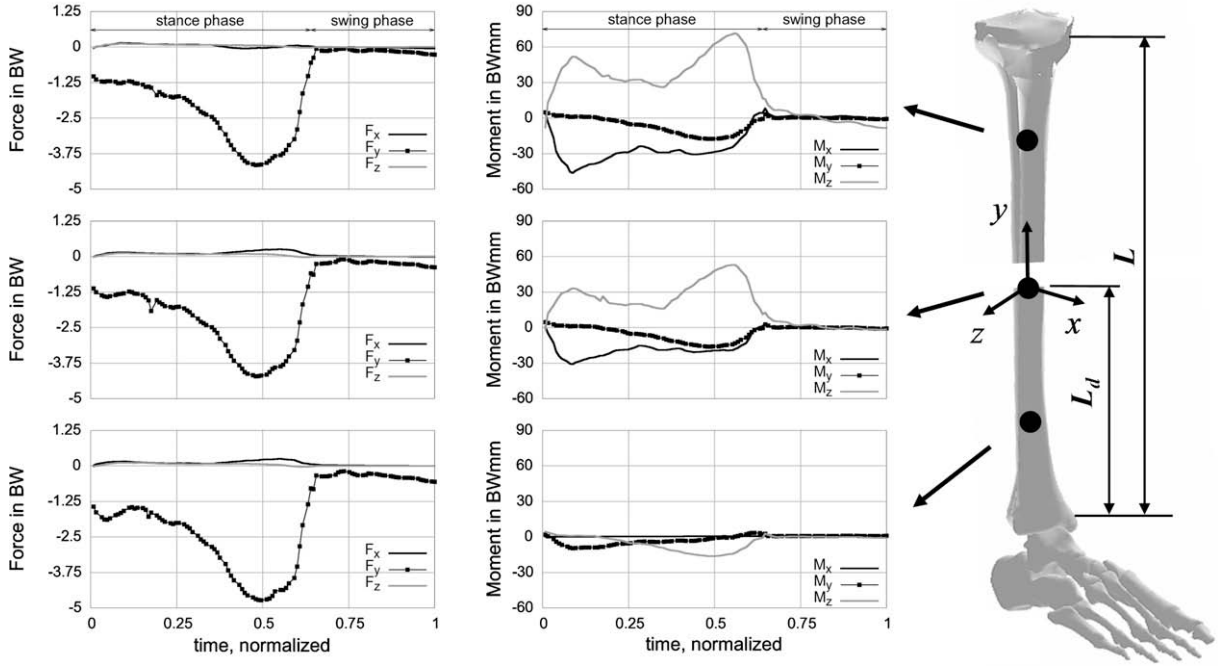


Fig. 3. Internal forces (left) and internal moments (middle) over the gait cycle at three locations along the tibial axis. The forces are given in bodyweight (BW) and the moments in bodyweight times millimetre (BWmm). Directions and location of the internal loads (right). The directions of the moments were according to the corkscrew rule.

Table 1
Comparision of the calculated peak loads of the resultant hip contact force and the knee contact force (axial force on the tibial plateau) to published in vivo data. The forces are given in bodyweight (BW).

Source	Hip contact force in BW	Knee contact force in BW
Calculation	4.4	3.3
Davy et al. (1988)	2.8	–
Bergmann et al. (1993)	4.1	–
Brand et al. (1994)	3.3	–
Bergmann et al. (2001)	2.9	–
Lu et al. (1997)	–	3.2
Lu et al. (1998)	–	2.8
Taylor et al. (1998)	–	2.5
Taylor and Walker (2001)	–	2.8
D'Lima et al. (2005)	–	2.8

data from telemetrizised total joint replacements. Qualitatively, both reaction forces showed loading peaks at the beginning and the end of the stance phase (Fig. 2) according to the results of the Julius–Wolff-Institut in Berlin (www.orthoload.com). The peak values of these forces (Table 1) determined in vivo in various studies ranged from 2.8 up to 4.1 BW for the resultant hip contact force and 2.5 up to 3.2 BW for the axial load on the tibial plateau. Based on these data, the calculated contact forces (4.4 BW and 3.3 BW) tend to be overestimated, however still in a physiological range. Therefore, we assume that the musculoskeletal model represents physiological internal forces and moments in the human tibia during gait.

Considering these limitations and assuming the normal gait as a representative loading scenario for in vitro testing, the supplementary data of this study could be used as follows. For static uniaxial tests of osteosynthesis implants, the peak loads of the internal loads at the location of the fracture could be used. For applying combined loading cases, the internal loads at the fracture location should be taken at that specific time point, when the internal loads seems to be most critical for the bone–implant complex. For testing e.g. the fatigue behaviour of the bone–implant complex, the internal loads at the fracture location of the whole gait cycle should be applied dynamically.

5. Conclusions

This study provides for the first time all internal forces and moments along the human tibial axis during gait. The results of this study could be used to define physiological loading scenarios for in vitro testing of osteosynthesis implants with respect to the location of the fracture along the tibial axis. Therefore, the mechanical behaviour of current implant designs could be optimized and implant failures in vivo might be avoided.

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Appendix A. Supplementary data

This text file includes all six internal loads over the gait cycle at nine different locations along the tibial axis. The location is defined by the distance to the ankle joint L_d in relation to the total length of the tibia L. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.clinbiomech.2008.12.007.

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